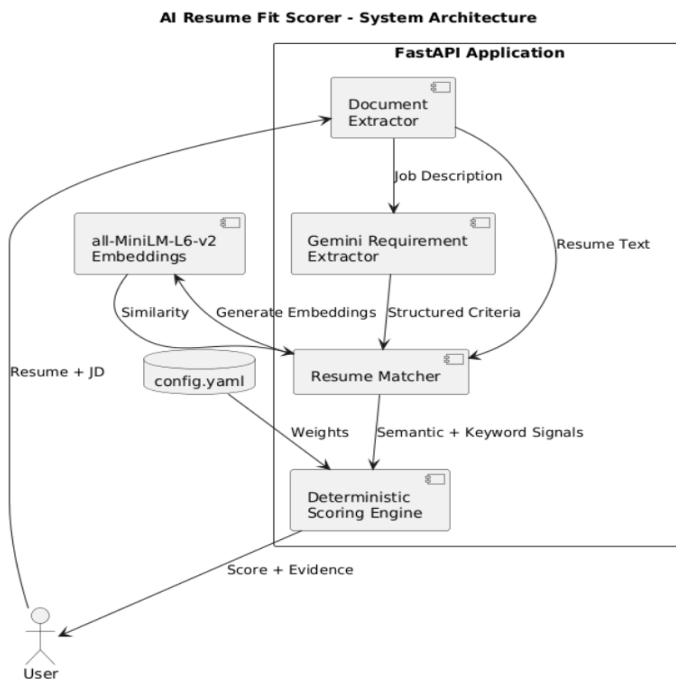


AI Resume Fit Scorer — Technical Explanation

An explainable resume-to-job matching system that extracts job requirements, evaluates resume evidence using semantic similarity and deterministic keyword matching, and produces a transparent 0–100 candidate-fit score.

1. **Objective :** The objective is to automatically compare a resume with a job description and generate an explainable fit score. The system extracts structured job requirements, matches them against resume content, combines semantic and keyword evidence, and produces deterministic criterion-level and overall scores.

2. System Architecture :



3. Processing Pipeline :

1. **Document Extraction:** PDF/TXT resume text is extracted and validated.
2. **Requirement Extraction:** Gemini converts the job description into structured criteria.
3. **Resume Matching:** Resume chunks are compared with criteria using semantic similarity and keyword evidence.
4. **Deterministic Scoring:** Criterion and category scores are calculated using configured weights.
5. **API Response:** FastAPI returns scores, matched keywords, evidence, and explanations.

4. Scoring Method :

Semantic similarity: all-MiniLM-L6-v2 generates embeddings for resume chunks and criteria. Cosine similarity provides the semantic signal.

Keyword evidence: Deterministic, case-insensitive matching identifies criterion keywords in the resume.

Keyword Score

$$\text{Keyword Score} = \frac{\text{Matched Keywords}}{\text{Total Criterion Keywords}}$$

Criterion Score

$$\text{Criterion Score} = (0.60 \times \text{Semantic Score} + 0.40 \times \text{Keyword Score}) \times 100$$

Category weights: Technical Skills 35%, Experience 25%, Responsibilities 20%, Education 10%, Tools 10%. Missing categories are proportionally renormalized.

5. Calibration Results :

Resume	Profile	Score
A	Senior Backend AI Engineer	66.6
B	Senior Backend AI Engineer	59.9
C	Junior Frontend / Designer	14.2

Two similar candidates produced an observed score gap of 6.7 points ($|66.6 - 59.9|$), compared with the configured maximum similar-resume gap of 15 points.

6. API & Validation :

API: GET /health and POST /api/v1/score. The scoring endpoint accepts a job description and PDF/TXT resume and returns the overall score, criterion scores, matched keywords, evidence, category scores, and deterministic reasons.

Testing: The project includes tests for document extraction, Gemini criteria extraction, API failures/retries, semantic matching, keyword matching, deterministic scoring, calibration, and FastAPI endpoints. Latest verification: **101 tests passed**.

7. Technology Stack :

Python • FastAPI • Google Gemini • Pydantic • pypdf • Sentence Transformers • all-MiniLM-L6-v2 • NumPy • PyYAML • Pytest